EASY · ADVANCED MATH · ANSWER KEY

Advanced Math

03

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 D

D. 12

Choice D is correct. Adding 53 to each side of the given equation yields $k^2 = 144$. Taking the square root of each side of this equation yields $k = \pm 12$. Therefore, the positive solution to the given equation is 12.

Choice A is incorrect. This is the positive solution to the equation $k^2 - 53 = 20{,}683$, not $k^2 - 53 = 91$.

Choice B is incorrect. This is the positive solution to the equation $k^2 - 53 = 5{,}131$, not $k^2 - 53 = 91$.

Choice C is incorrect. This is the positive solution to the equation $k^2 - 53 = 1{,}391$, not $k^2 - 53 = 91$.

Q2 A

A. $5x^2 - 2x + 3$

Choice A is correct. The given expression $(2x^2 - 4) - (-3x^2 + 2x - 7)$ can be rewritten as $2x^2 - 4 + 3x^2 - 2x + 7$. Combining like terms yields $5x^2 - 2x + 3$.

Choices B, C, and D are incorrect and may be the result of errors when applying the distributive property.

Q3

B

B. At the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice B is correct. The y-intercept of a graph in the xy-plane is the point where $x = 0$. For the given equation, the y-intercept of the graph in the xy-plane can be found by substituting 0 for x in the equation, which yields $y = -\frac{1}{4}(0)^2 + 20 + 29$, or $y = 29$. Therefore, the y-intercept of the graph is $(0, 29)$. It's given that y is the estimated number of scheduled deliveries x months after the end of May 2012. Therefore, $x = 0$ represents 0 months after the end of May 2012, or the end of May 2012. Thus, the best interpretation of the y-intercept of the graph of this equation in the xy-plane is that at the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q4

C

C. $-\frac{7}{2}, \frac{9}{2}$

Choice C is correct. The solution to a system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at a point with an x-coordinate between -4 and -3 and a y-coordinate between 4 and 5. Of the given choices, only $-\frac{7}{2}, \frac{9}{2}$ has an x-coordinate between -4 and -3 and a y-coordinate between 4 and 5.

Choice A is incorrect. This is the y-intercept of the graph of the linear function.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Q5

D

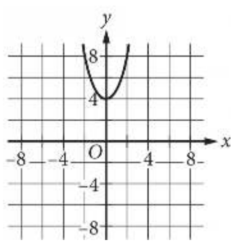
D. $r^3+6r^2+8r+19$

Choice D is correct. Grouping like terms, the given expressions can be rewritten as $r^3+(5r^2+r^2)+8r+(7+12)$. This can be rewritten as $r^3+6r^2+8r+19$.

Choice A is incorrect and may result from adding the two sets of unlike terms, r^3 and r^2 as well as $5r^2$ and $8r$, and then adding the respective exponents. Choice B is incorrect and may result from adding the unlike terms r^3 and r^2 as if they were r^3 and r^3 and adding the unlike terms $5r^2$ and $8r$ as if they were $5r^2$ and $8r^2$. Choice C is incorrect and may result from errors when combining like terms.

Q6

D

D. 

Choice D is correct. For the quadratic function $f(x)=x^2+4$, the vertex of the graph is $(0,4)$, and because the x^2 term is positive, the vertex is the minimum of the function. Choice D is the only option that meets these conditions.

Choices A, B, and C are incorrect. The vertex of each of these graphs doesn't correspond to the minimum of the given function.

Q7

B

B. $Q = \frac{T}{P}$

Choice B is correct. Solving the given equation for Q gives the quantity of the product sold in terms of P and T. Dividing both sides of the given equation by P yields $\frac{T}{P} = Q$, or $Q = \frac{T}{P}$. Therefore, $Q = \frac{T}{P}$ gives the quantity of product sold in terms of P and T.

Choice A is incorrect and may result from an error when dividing both sides of the given equation by P. Choice C is incorrect and may result from multiplying, rather than dividing, both sides of the given equation by P. Choice D is incorrect and may result from subtracting P from both sides of the equation rather than dividing both sides by P.

Q8

B

B. $2xy8x^2y + 7$

Choice B is correct. Since $2xy$ is a common factor of each term in the given expression, the expression can be rewritten as $2xy8x^2y + 7$.

Choice A is incorrect. This expression is equivalent to $16x^2y^2 + 14xy$.

Choice C is incorrect. This expression is equivalent to $28x^3y^2 + 14xy$.

Choice D is incorrect. This expression is equivalent to $112x^3y^2 + 14xy$.

Q9 D

D. 320

Choice D is correct. It's given that the function f is defined by $fx = 5x^2$. Substituting 8 for x in $fx = 5x^2$ yields $f8 = 58^2$, which is equivalent to $f8 = 564$, or $f8 = 320$. Therefore, the value of $f8$ is 320.

Choice A is incorrect. This is the value of $f8$ if $fx = 5x$.

Choice B is incorrect. This is the value of $f8$ if $fx = 5x + 2$.

Choice C is incorrect. This is the value of $f8$ if $fx = 5x2$.

Q10 B

B. $-2, 6$

Choice B is correct. The solution x, y to the system of two equations corresponds to the point where the graphs of the equations intersect in the xy-plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point $-2, 6$. Thus, the solution x, y to this system is $-2, 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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